

Adaptive Diffusion Priors on Pre-clinical CT Reconstruction

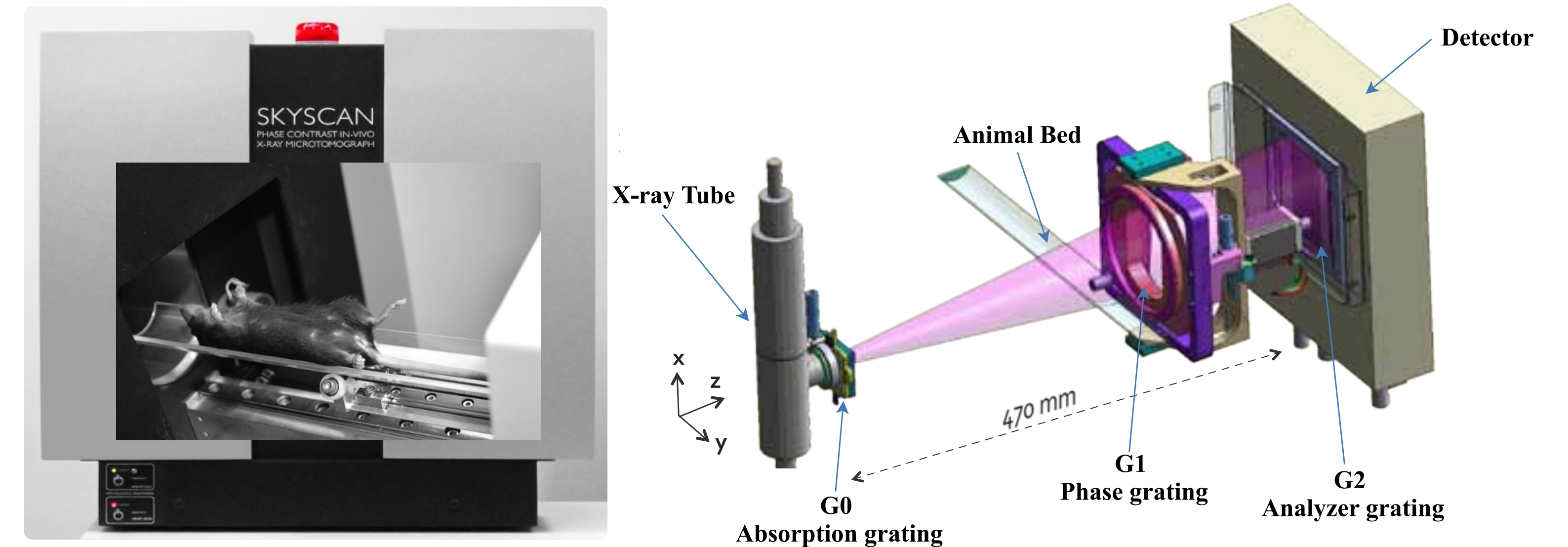
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INTRODUCTION

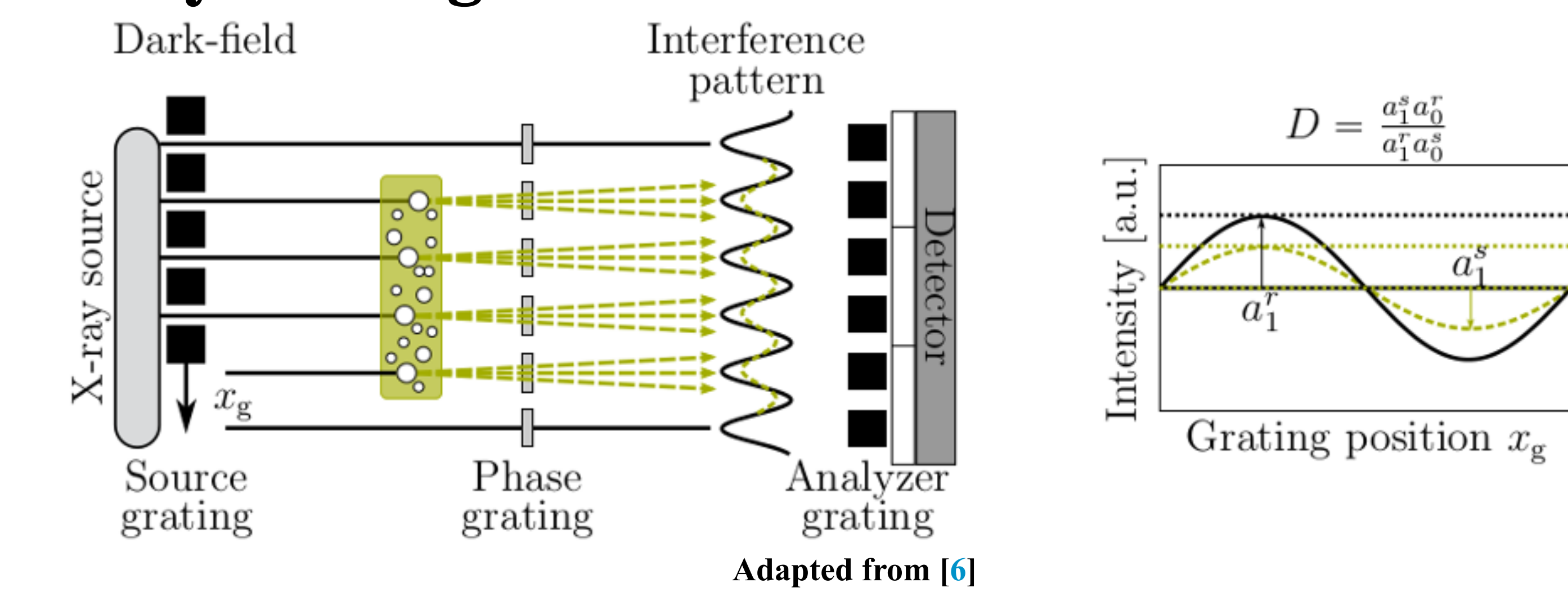
- X-ray dark-field imaging enables visualization of **lung microstructure** and detection of **pulmonary diseases** [1,2,3]
- Reconstruction suffers from **streak artifacts** due to highly **undersampled acquisitions** constrained by radiation dose.
- **Diffusion-based reconstruction**
 - Adaptation of Deep Diffusion Image Prior (DDIP) [4] to Dark-Field CT.



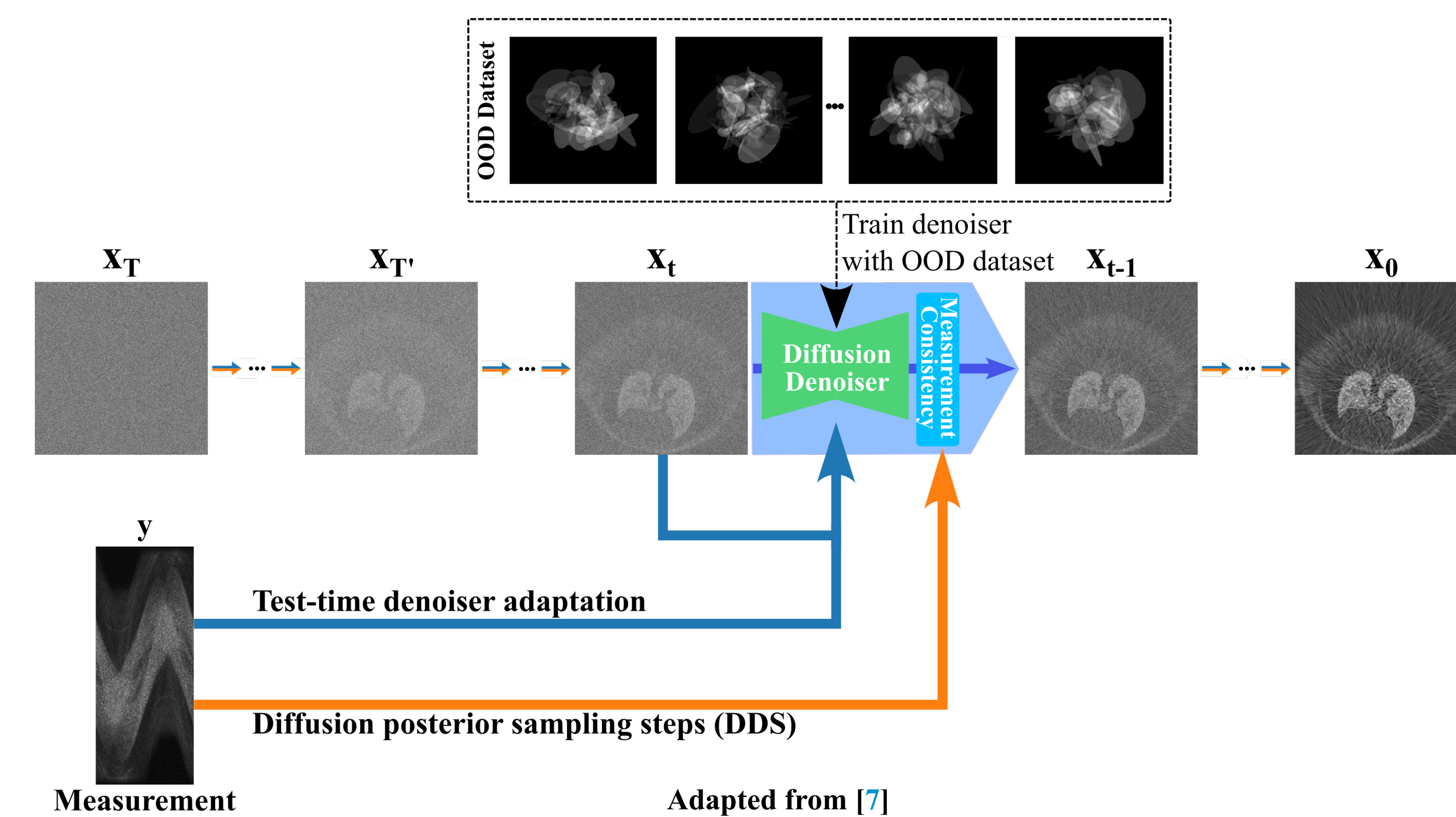
Skyscan 1190. Adapted from [5]

METHOD

X-ray Grating Interferometer



Adaptive Diffusion Prior

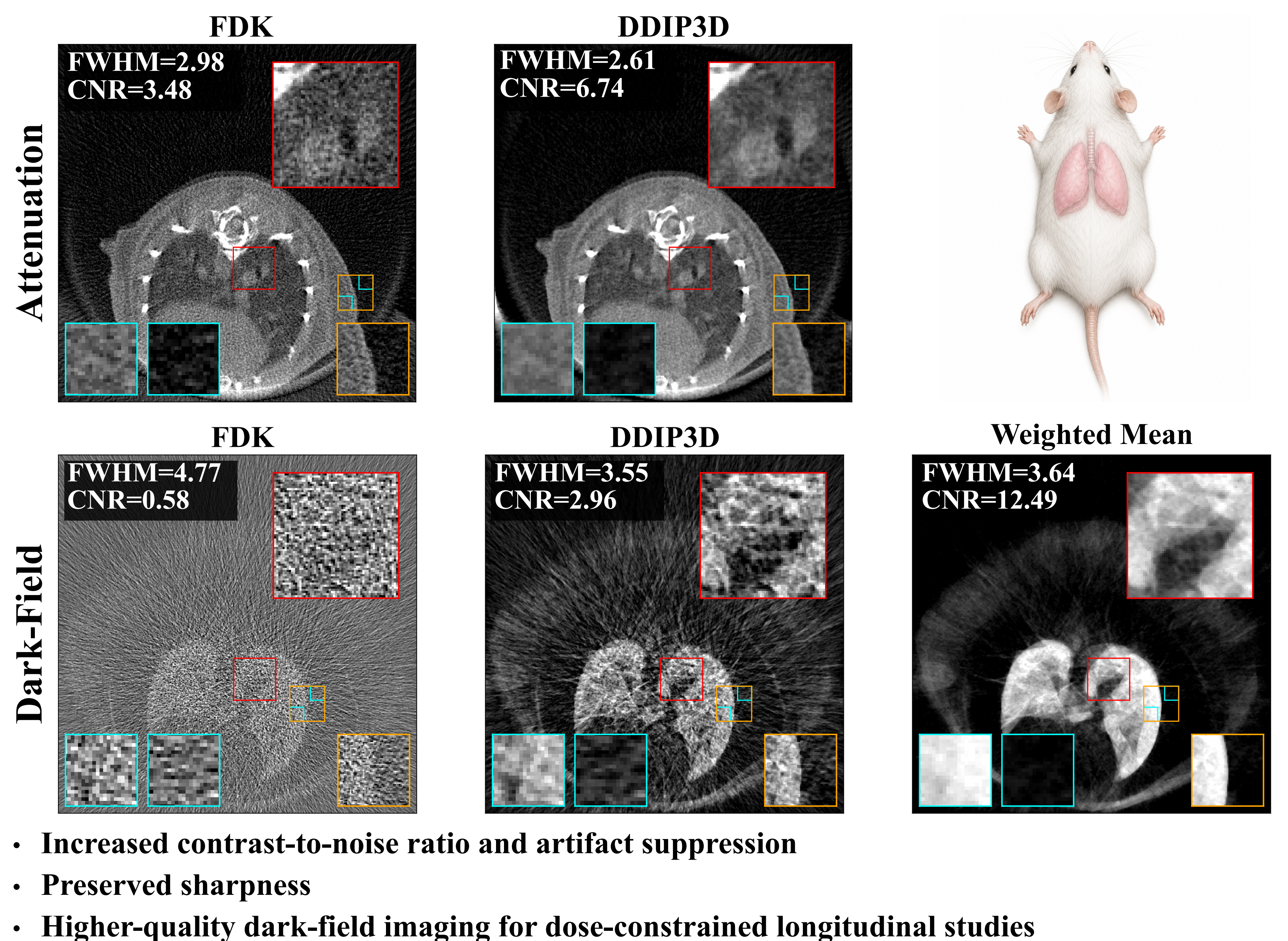


- Pretrained diffusion model as an **image prior**
- TIGRE-based [8] forward and backprojected operators for **physics-based consistency**
- Alternative method of multipliers (ADMM) [9] with **total variation (TV) regularization** to enforce **data consistency**
- Low-rank adaptation (LoRA) [10] for adaptation to **measured data**

Weighted Mean

- Aggregates intermediate diffusion estimates along the **reverse trajectory**
- Combines **coarse global structure** from **early diffusion stages** with **fine details** from **later stages**
- Suppresses **high-frequency artifacts** and **preserves anatomical consistency**

RESULTS



CONCLUSION

- Integrated diffusion models as **learned priors** with **physics-based reconstruction** and **out-of-distribution adaptation** for addressing the undersampling problem in **dark-field CT**
- Future work - reader studies to assess the **clinical impact** of the proposed method

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Financial support through the European Research Council (ERC Synergy Grant SmartX, SyG 101167328), and the Free State of Bavaria under the Excellence Strategy of the Federal Government and the States, as well as by the Technical University of Munich - Institute of Advanced Study.